

Bandit - Level 08

Goal

The password for the next level is stored in `data.txt`, on the **only line that occurs once**. Every other line in the file is repeated.

Connection

```
ssh -p 2220 bandit8@bandit.labs.overthewire.org
```

Use the password retrieved at the end of [Bandit - Level 07](#).

Concepts

`uniq` does **not** search the whole file for duplicates. It only compares each line against the **Immediately preceding one**. Lines that are identical but sit far apart in the file are invisible to it.

That single behavior is why `sort` is not decoration here — it's a **precondition**. Sorting groups identical lines into adjacent blocks so that `uniq` can actually see them as repeats.

Flag	Effect
<code>-u</code>	print only lines that appear exactly once
<code>-d</code>	print only lines that are repeated
<code>-c</code>	prefix each line with its occurrence count
<code>-i</code>	ignore case when comparing

Full stream/pipe mechanics: [Linux - Piping and Redirection](#).

Attempts

```
# Reflex from previous levels - reached for find to locate the file first
find / data.txt
# → dumped the entire filesystem and errored out.
# "data.txt" without -name is parsed as a second STARTING PATH, not a search
criterion.

# The file was in the home directory all along
pwd                # /home/bandit8
cat data.txt       # printed 1001 lines - readable, but far too many to scan by eye
```

Then, before solving, three commands run deliberately to **prove the** `uniq` adjacency behavior instead of taking it on faith:

```
wc -l data.txt
# → 1001 data.txt          total lines in the file
```

```
uniq -u data.txt | wc -l
```

```
# → 981
```

WRONG. Without sorting, `uniq` only catches adjacent repeats, so 981 lines falsely look "unique"

```
sort data.txt | uniq -c | head
```

```
# → 10 0LTDNpAmqqfuE0FLE0ksGF6c0Kraspzs
```

```
# 10 1cKKjk7M0PL2cPubYgc9W4307bYC0ohF
```

```
# 10 1PesxCa7cihwvCvzBeKAcjKkjUwp7i2z
```

```
# ... every line appears exactly 10 times
```

The arithmetic closes cleanly: **100 distinct lines × 10 repetitions = 1000, plus 1 unique line = 1001**. The structure of the file was fully mapped before the answer was extracted.

Solution

```
sort data.txt | uniq -u
```

```
# → Ejm0SvuAu7sGAHqHVcBDPirRe9T03kxL
```

`sort` groups the identical lines into adjacent blocks; `uniq -u` then discards every block with more than one member, leaving the single orphan line.

Points of Friction

1. Reached for `find` again to locate a file that was in the home directory. Same friction as [Bandit - Level 07](#) — second consecutive level. In Bandit, unless the level text explicitly points elsewhere (as Level 5 does with `inhere/`), the target file is in `~`. New standing rule: `pwd` and `ls -la` before any `find`. Cheap reconnaissance first.
2. `find / data.txt` is not a search. `find`'s syntax is `find [starting-paths...] [tests] [actions]`. Any bare word that doesn't begin with `-` is consumed as a **starting path**, so this asked `find` to recurse through `/` and through a nonexistent directory named `data.txt`. The criterion only exists once `-name` is present: `find / -name data.txt 2>/dev/null`. Detailed in [Linux - find Command](#).
3. `uniq -u` alone looked like it worked. It returned output, no error, 981 lines. A command that runs successfully is not a command that answered the question — the same trap already recorded in [Linux - File Type Detection](#), in a different disguise.

Key Takeaway

`uniq` is a **line-adjacency filter**, not a deduplicator. It has no memory of the file beyond the previous line, which is what makes it fast enough to stream gigabytes — and what makes `sort` mandatory in front of it whenever duplicates aren't already grouped.

The pipeline `sort | uniq -c | sort -rn` is the single most reused pattern in log analysis: it turns a raw stream of events into a frequency table ranked by volume. `uniq -u` is its mirror image — instead of finding what happens most, it finds **what happened only once**, which in a security context is usually the interesting part. Full treatment: [Linux - Sorting and Deduplication](#).

Next

```
ssh -p 2220 bandit9@bandit.labs.overthewire.org
```

